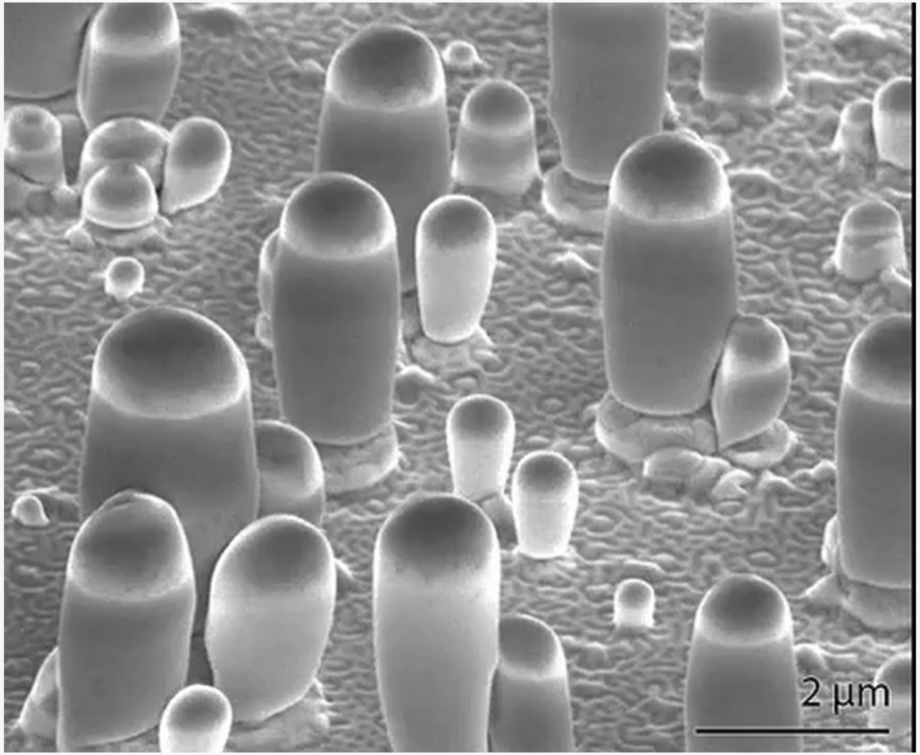
Spontaneous Growth of Gallium-Filled Microcapillaries on Ion Bombarded GaN

Botman, Bahm, Randolph, Straw, and Toth

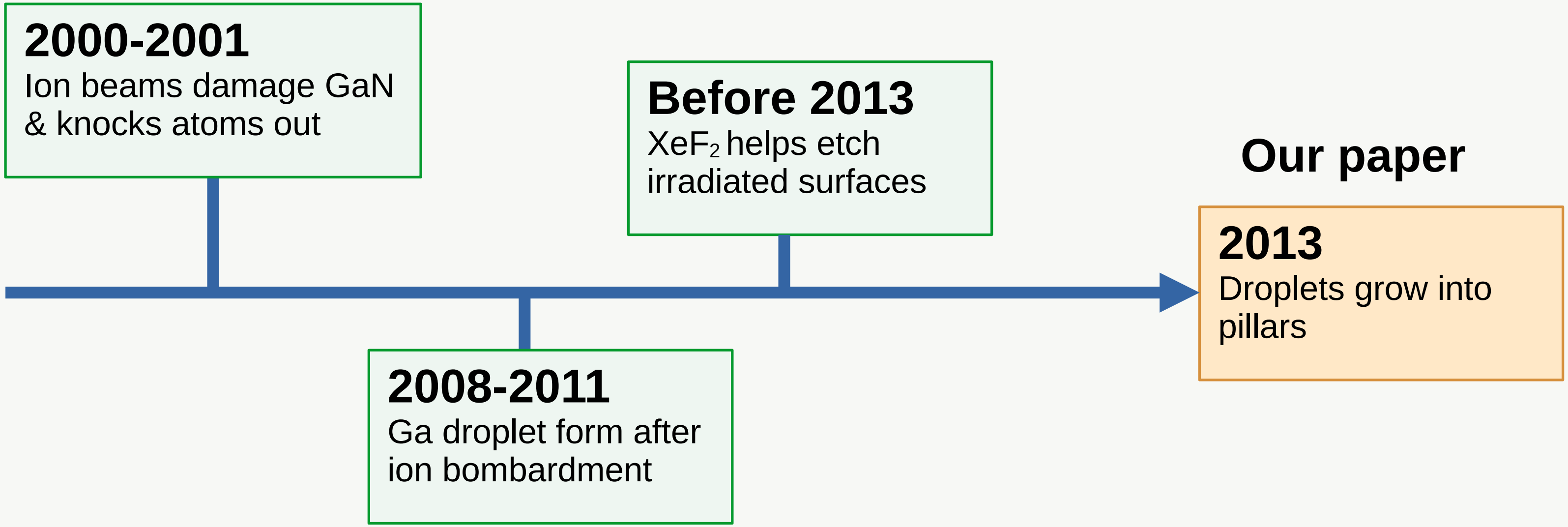
Physical Review Letters, 2013



Presented by: Ekvin

Why was this paper surprising?

The ingredients were known separately, but nobody expected them to grow pillars together.



A tool used for etching suddenly behaves like a growth process.

Experimental Setup

- GaN substrate
- Ga^+ focused ion beam
- XeF_2 gas injector
- SEM / electron beam

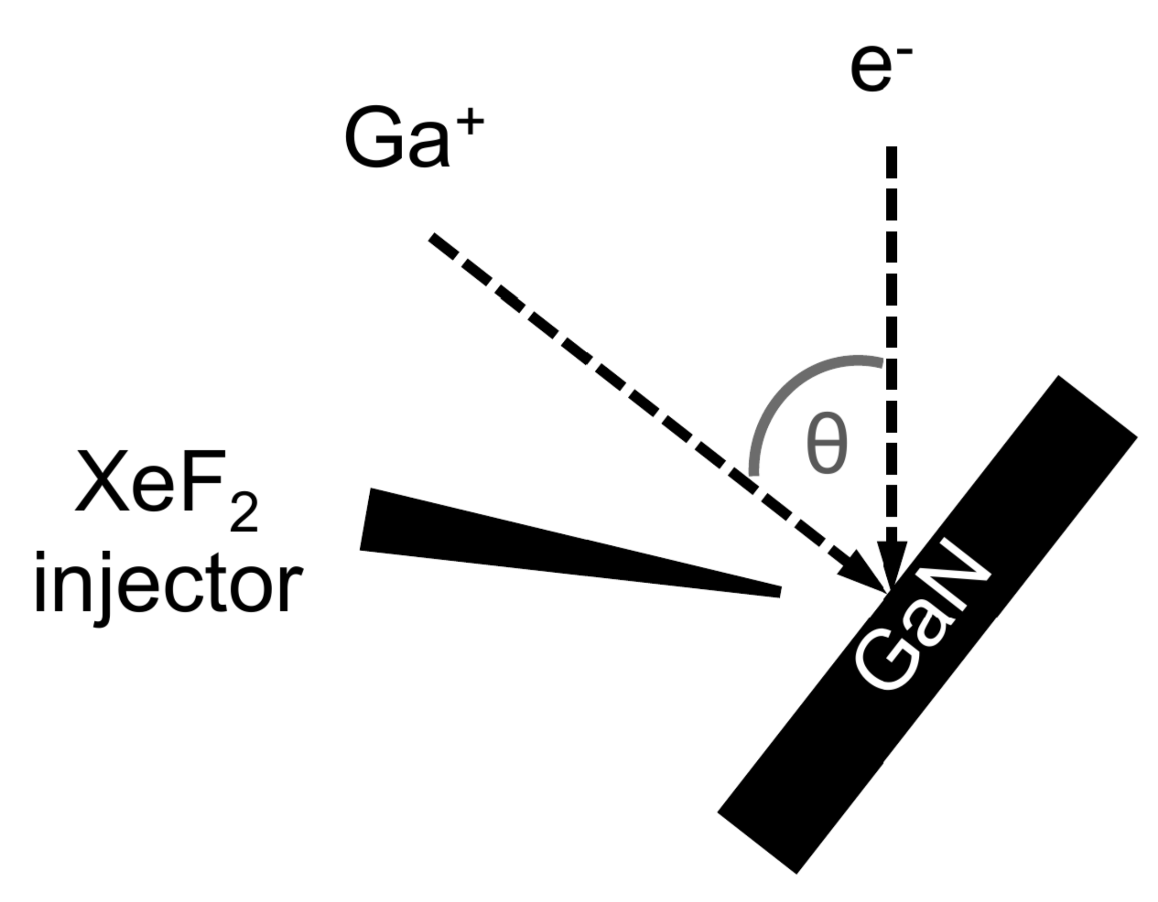


Fig. S1: Ga+ beam, electron beam, XeF2 gas injector, and GaN substrate.

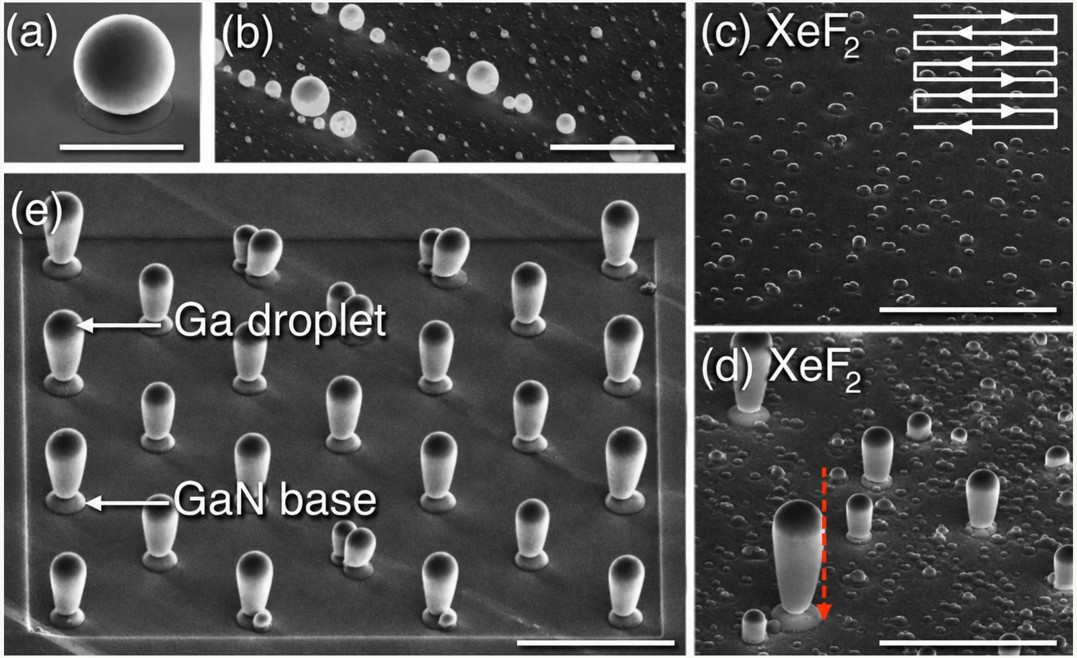
The Discovery: *XeF_2 Turns Droplets into Pillars*

Without *XeF_2*

Ga^+ irradiation of GaN forms Ga droplets

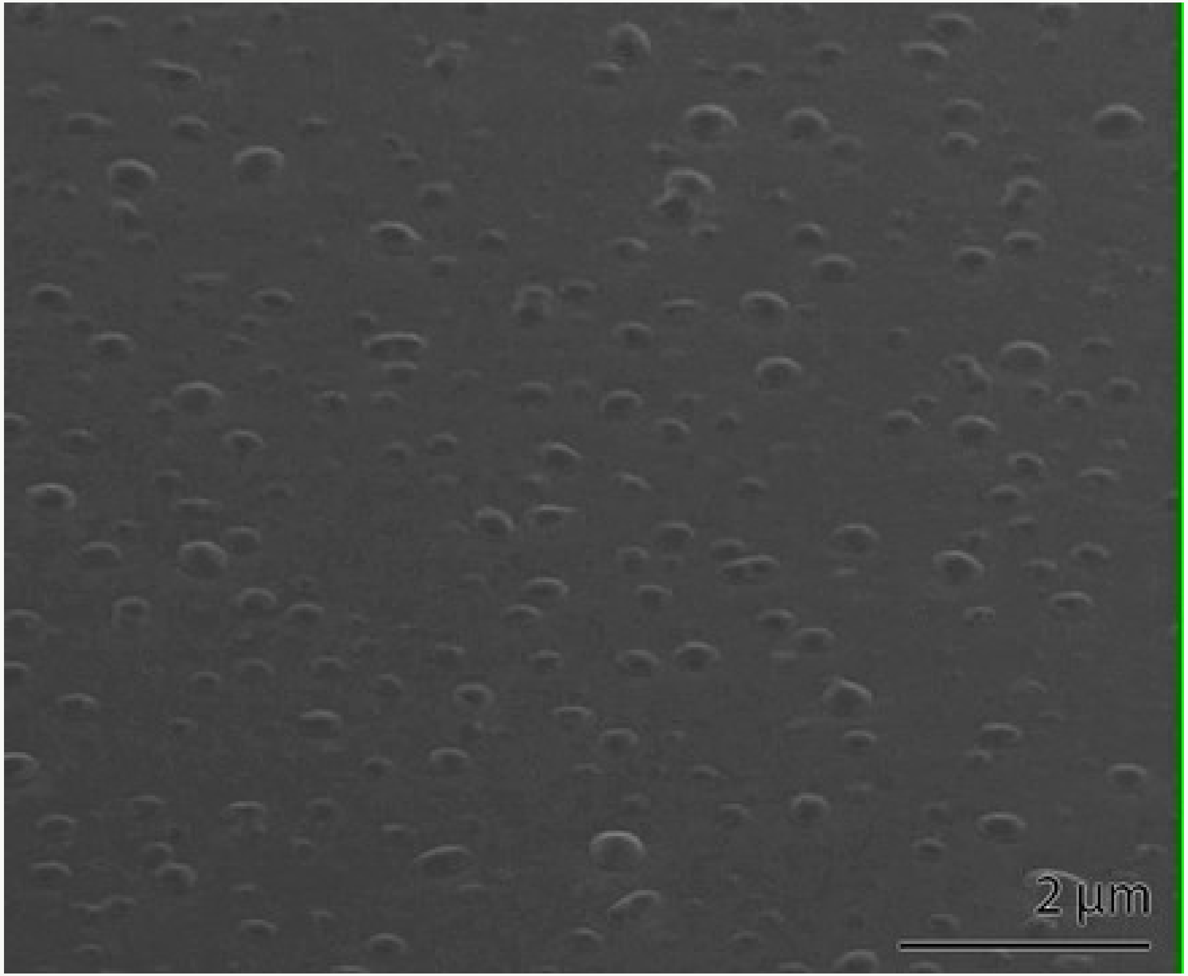
With *XeF_2*

- Droplets become starting points for vertical pillars
- Ordered arrays can be made by FIB pre-patterning



Key surprise: *A normally destructive beam/etching process produces growth.*

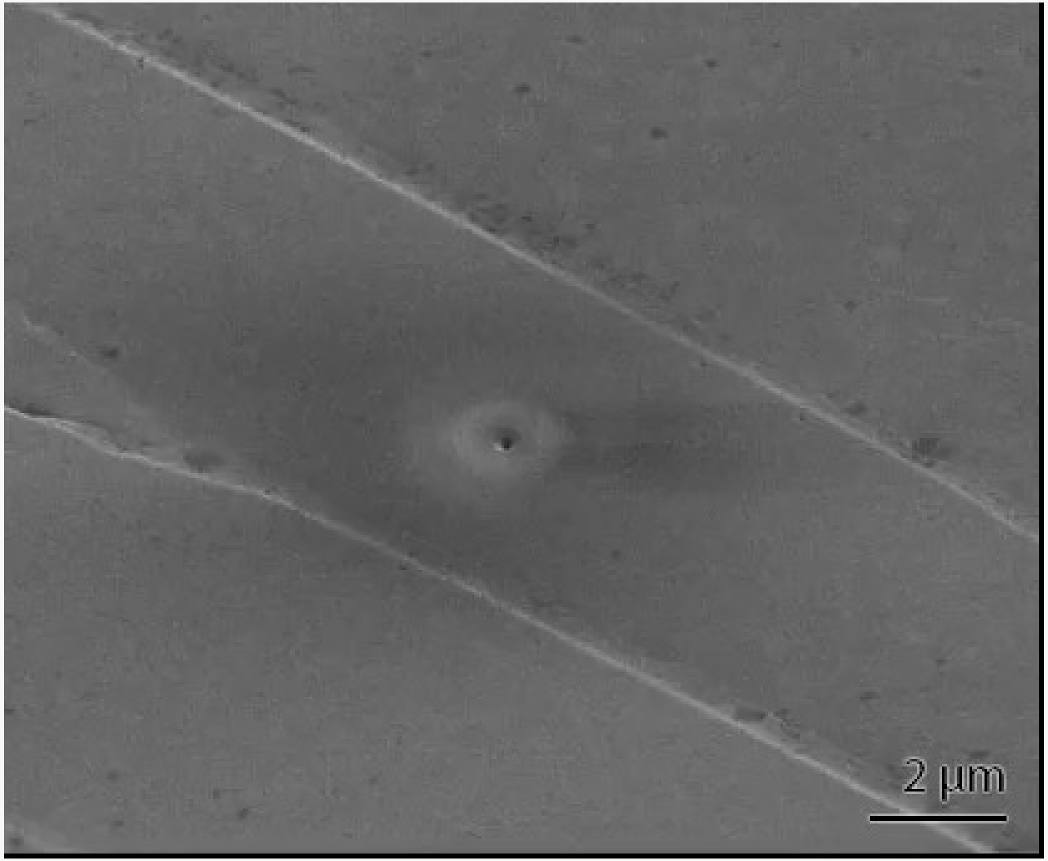
Watching Pillars Sprout



Supplemental Video 1

Many droplets rise into pillars during repeated FIB scanning

Growth Process: Droplet to Pillar to Stop



Supplemental Video 2

What to notice:

- A droplet nucleates at a small pit or defect.
- The droplet rises as a pillar forms underneath.
- Growth slows and eventually stops by itself.

Same beam scan:

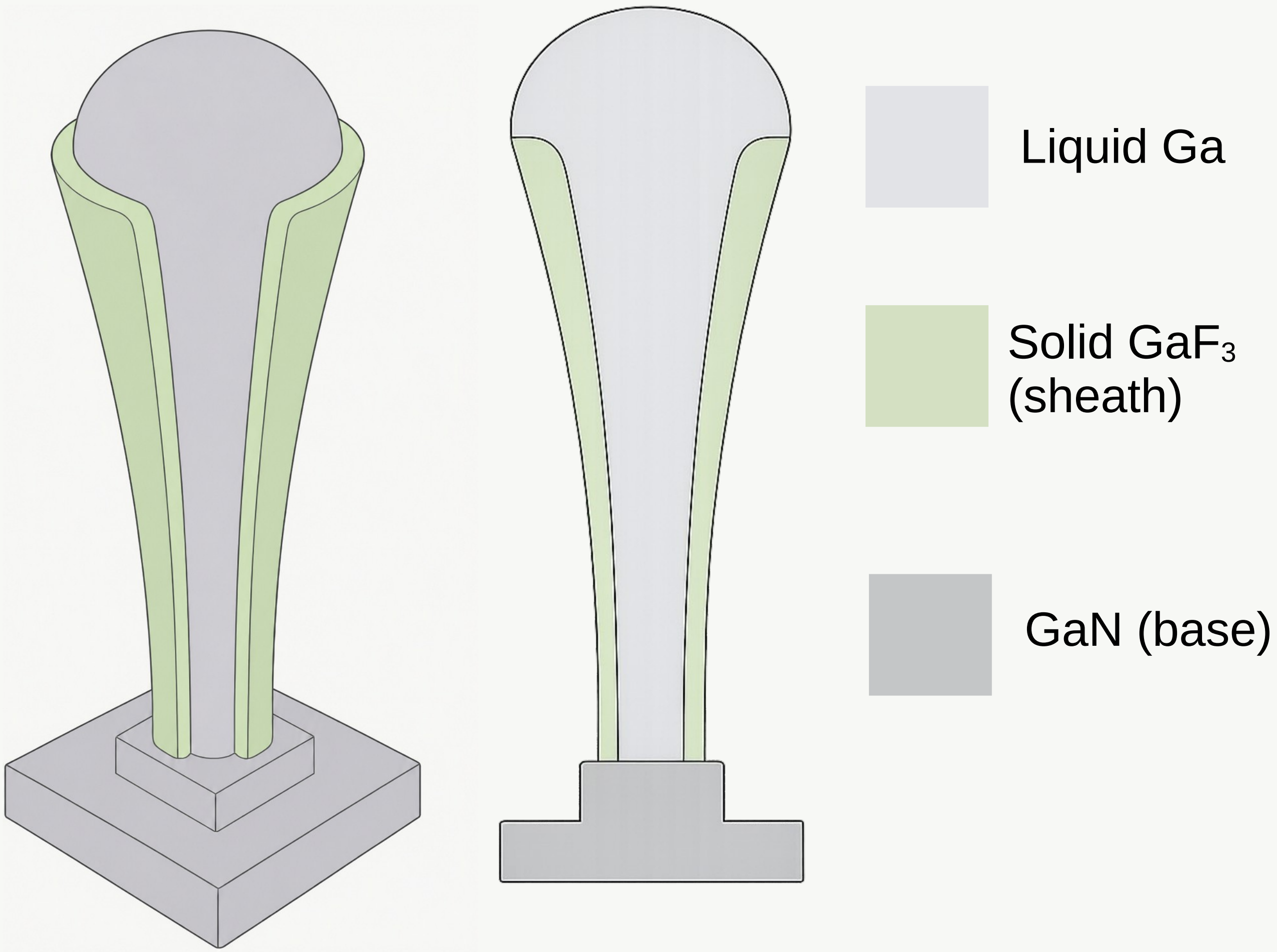
starts growth, feeds it,

then makes it stop.

What Is the Pillar Made Of?

Simplified schematic:

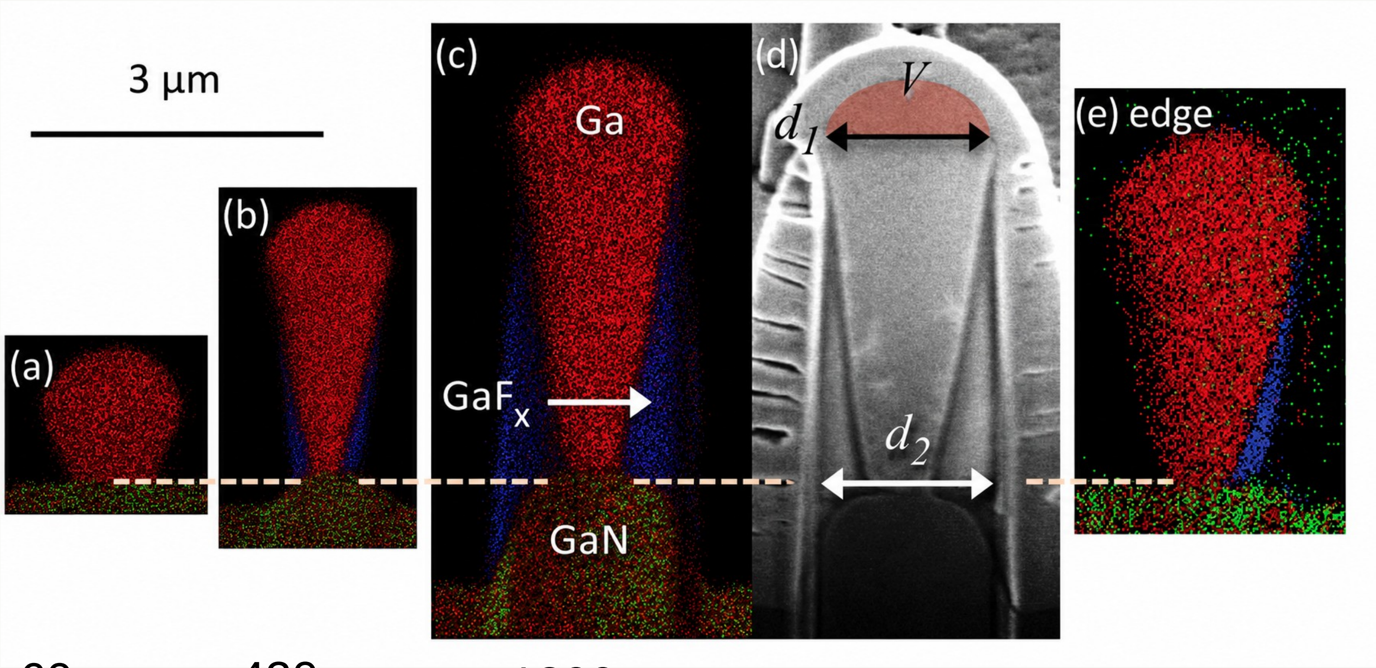
Liquid Ga inside
Solid GaF_3 outside
GaN base underneath



EDS Proves the Core-Sheath Structure

EDS maps elements by color:

Red = Gallium
Blue = F (Fluorine)
Green = N (Nitrogen)



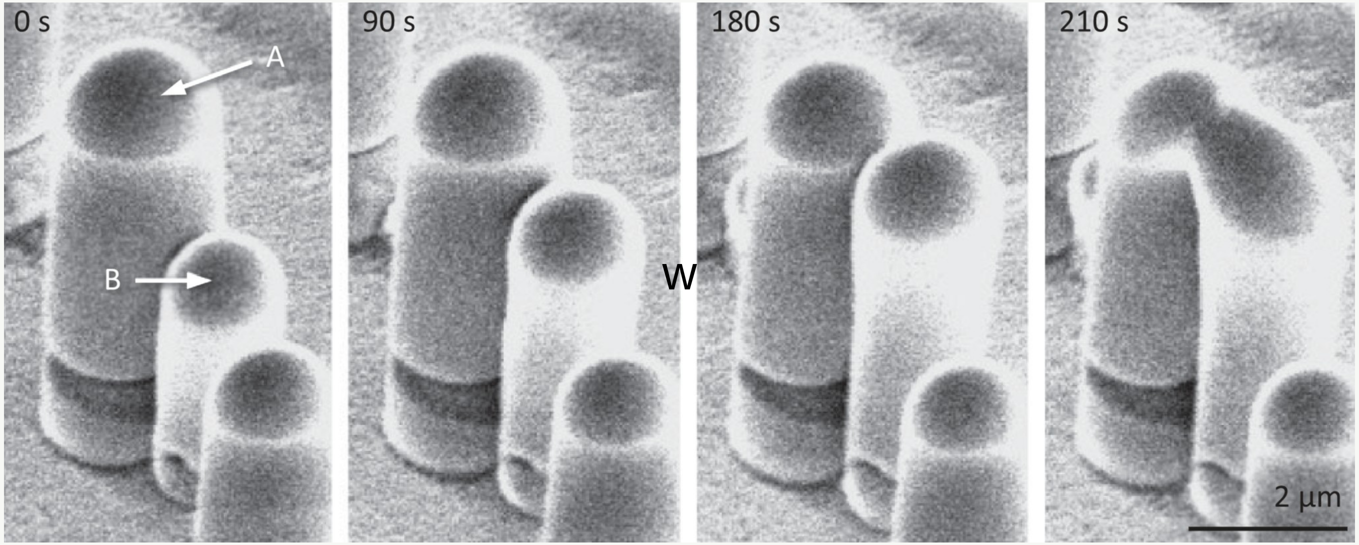
60 s
420s
1200s

Conclusion:
The pillar is Ga-filled, not carved GaN.

Proof of the Liquid Cap: Droplets Merge

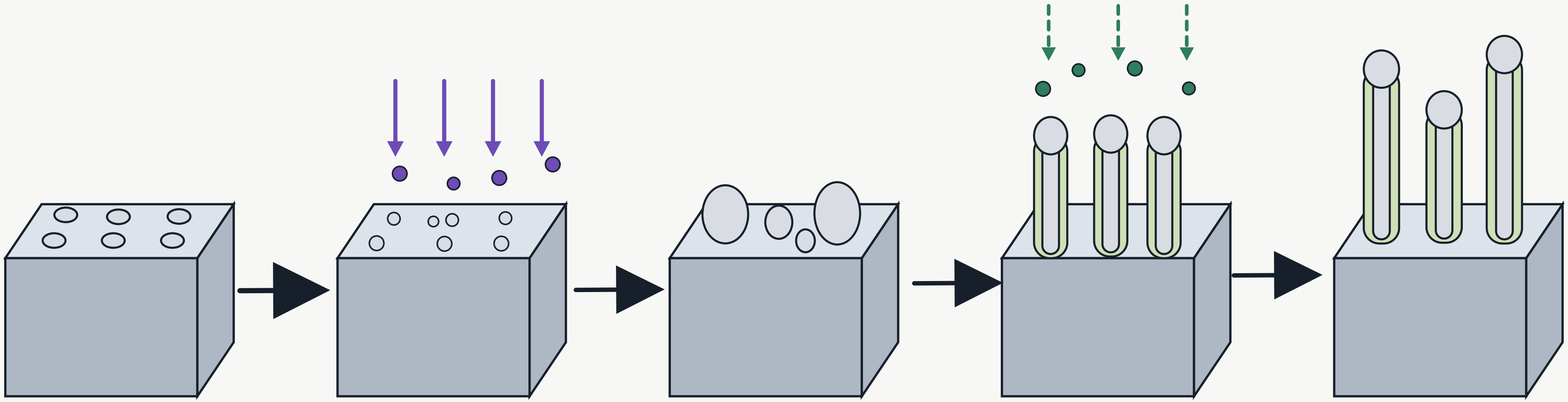
What This Proves:

- Ga caps behave like liquid droplets.
- When two caps touch, they merge.



Conclusion: The top cap is not a solid plug; it is liquid Ga.

Mechanism: Four Effects Reinforce Each Other



1. Beam breaks GaN
2. Ga pools into droplet
3. XeF_2 forms fluorinated wall
4. Ga cap protects growth

Liquid Ga

Solid GaF_3 (sheath)

GaN (base)

Ga^+ ions

XeF_2 ions

Not VLS: no catalyst particle; Ga is the material being shaped

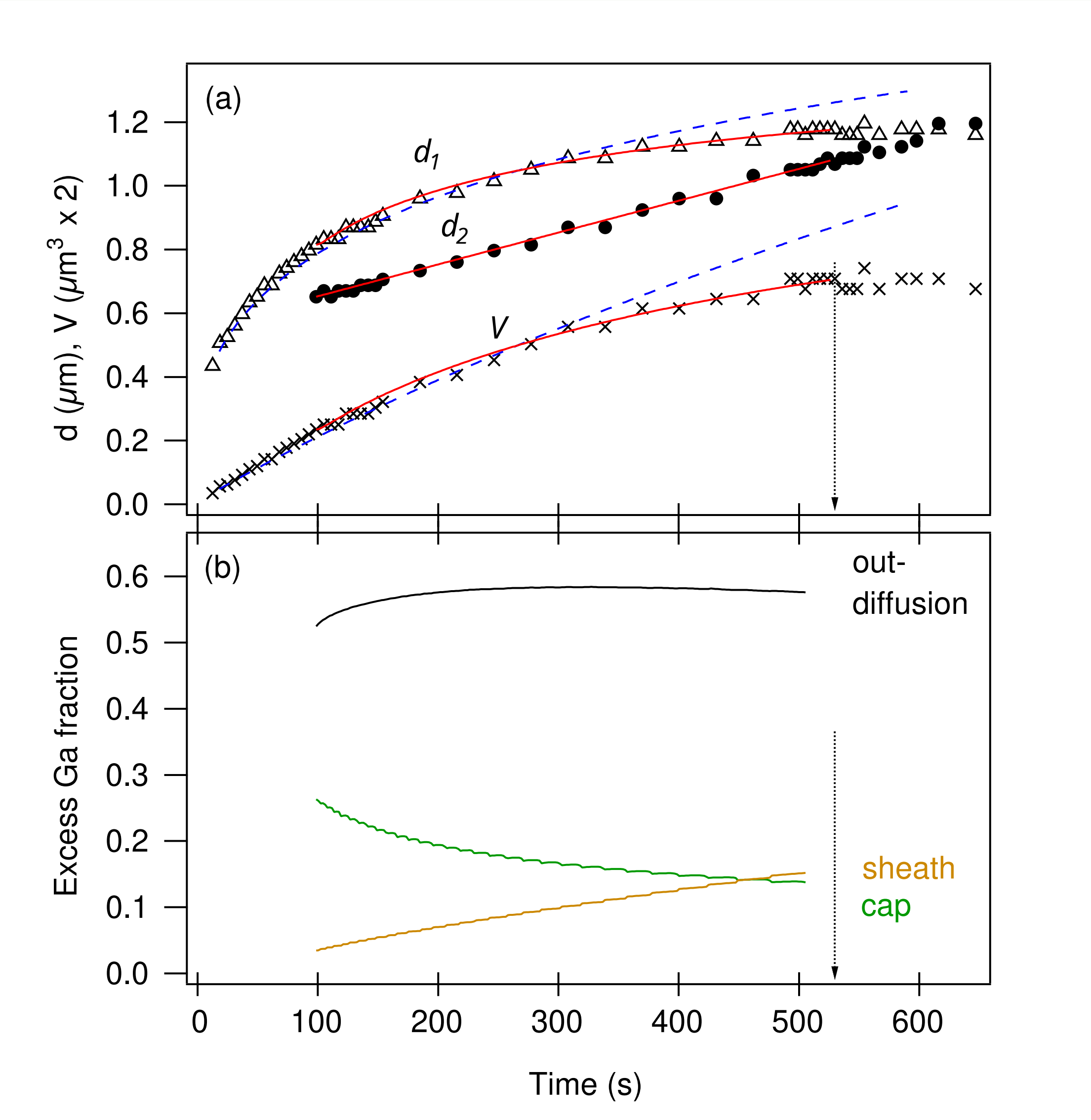
Why Growth Stops

Growth slows because Ga supply becomes limited.

Supply: Ga diffuses up from the substrate along the sheath

Loss: Ga becomes sheath or gets sputtered away

The pillar runs out of usable Ga at the cap



What Is Strong, and What Is Limited?

Strong evidence

- Live growth videos
- EDS proves core-sheath structure
- Growth curves show self-limiting behavior

Limits

- Rough height/diameter control
- Mechanism partly model-supported
- Proof of concept, not a device process

Outlook: Turning Beam Damage Into Growth

What It Shows:

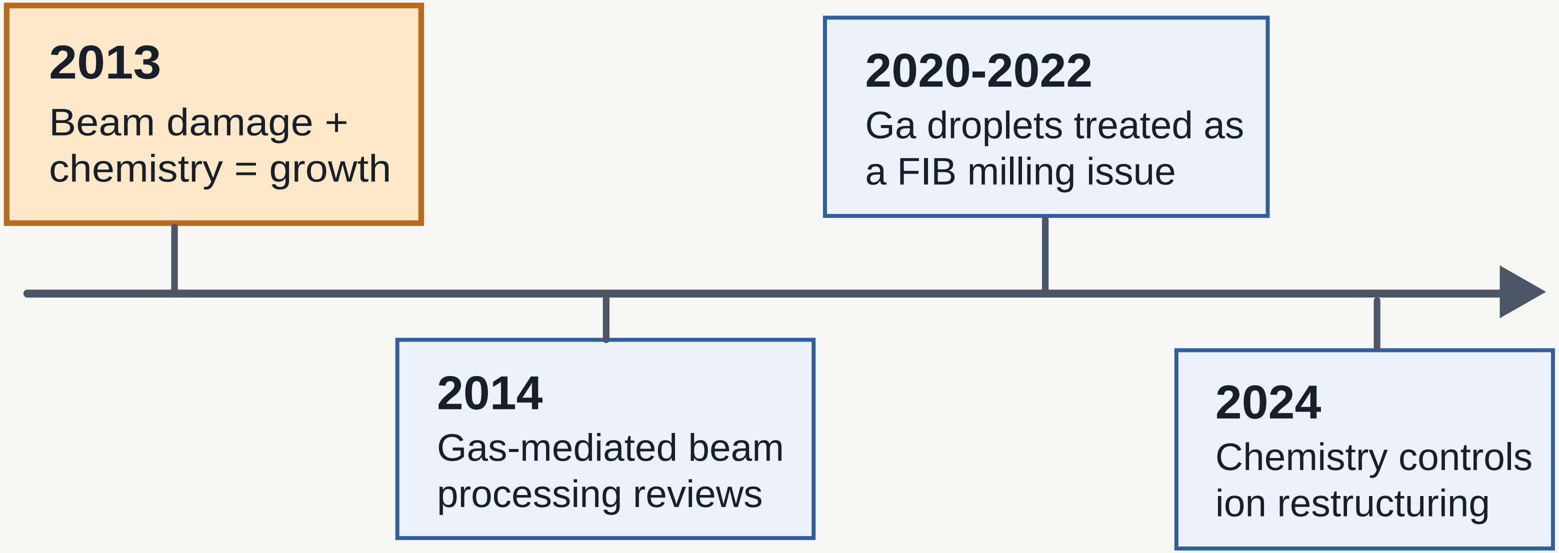
- FIB is not only subtractive
- XeF_2 chemistry turns Ga droplets into core-sheath pillars
- Growth requires generation + diffusion + self-masking.

What Remains Open:

- Predictable height/diameter control
- Testing other III-V materials
- Can it become a reliable fabrication process?

What did this paper influence?

It was not a huge device platform. Its influence is mainly mechanistic.



The impact is niche, but useful:

Final Takeaway

A normally destructive ion-beam process can become organized growth.

Ga left behind + XeF_2 chemistry + surface diffusion + self-masking

Questions?

References

1. Botman, A.; Bahm, A.; Randolph, S.; Straw, M.; Toth, M. Spontaneous Growth of Gallium-Filled Microcapillaries on Ion-Bombarded GaN. Physical Review Letters 111, 135503 (2013).

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Figures and videos from Botman et al. except for the cartoons.